

Pharos

A lighthouse for drug safety

Drug Safety Signal Detector & Regulatory Submission Readiness Checker

Team Sudo Salingers · IBM Bob AI Innovation Hackathon 2026 · Problem P2
(Pharma & Biotech)

The Pharos was the lighthouse of Alexandria — a beacon that warns of danger and guides ships safely into port. Two jobs, one light. That's the product.



The Problem — Who, What, Why It Hurts

20M+

Adverse-event reports
in FDA's FAERS. +1M a year.

27K+

Heart attacks

Vioxx: cardiovascular signal sat in the data; **27,000+ heart attacks** before withdrawal (2004).

100K

Page drug dossiers
assembled by hand in the 5-module ICH CTD structure.

\$50–100M

Cost of one missing section

One missing section → refuse-to-file → **6–12 months, \$50–100M.**

Same root cause: *too much structured data for manual review.*



Two people: the pharmacovigilance scientist who can't read the pile, and the regulatory lead who can't see the hole in the dossier until the agency does.

Why Nothing Fixes It Today

Manual review

doesn't scale past hundreds of reports

Commercial PV platforms

expensive, closed, periodic — not conversational

Notebook scripts

untested, irreproducible, unexplainable

Excel checklists

no link to the ICH structure, no weighting

Plain LLM chat

will confidently invent a PRR



Missing piece: a deterministic, tested evidence engine over *real* data, with an AI on top that *reads* the evidence.

The Solution — Two Modes, One Engine

Mode 1 · Signal Detection

Live openFDA → 2×2 table → **PRR · ROR (95% CI) · χ^2 · Evans 2001** → rank → filter admin noise → cluster by organ system.

Mode 2 · Submission Readiness

ICH M4 CTD as a **69-section machine-readable checklist** (required · weight · why-it-matters) → match outline → per-module score → ranked gaps.

IBM Bob = the analyst. Calls the tools over MCP, verifies, reads cases, writes the memo.

Pharos supplies numbers. Bob supplies judgement. Neither pretends to be the other.

It Finds Real Disasters in Real Data

`python -m pharos scan rofecoxib --alias vioxx --live` against 20.7M reports:

Reaction	n	PRR [95% CI]	χ^2
Coronary artery disease	4,156	72.2 [70.0–74.5]	251,805
Myocardial infarction	17,981	52.2 [51.5–52.8]	815,462
Cerebrovascular accident	13,370	39.1 [38.5–39.7]	459,992

Cardiac SOC cluster: **8 signals, 35,521 cases**. Same result for Avandia, Baycol, Bextra, Meridia, Darvon.

 *Honest framing: openFDA starts 2004, Vioxx withdrew Sept 2004, and litigation inflated these counts. This proves the method works on real data – nothing more. The next slide is the real claim.*

The Avandia Timeline — When Could We Have Known, and What Hid It?

```
pharos timeline rosiglitazone "myocardial infarction" --alias avandia -x rofecoxib -x vioxx --to 2013
```

Year	Avandia MI share	FAERS MI background	Vioxx share of ALL MI reports	Standard PRR	Vioxx-excluded PRR
2005	3.3%	4.5%	68%	0.73	2.13
2006	3.8%	5.5%	72%	0.68	2.27 •
2007	5.7%	2.5%	15%	2.26	2.66

 Standard screen first flags: 2008 <i>after</i> the Nissen meta-analysis (May 2007) and FDA's boxed warning (Nov 2007).	 Pharos masking check Vioxx litigation reports were ~70% of every MI report in FAERS — the background Avandia was compared against was inflated 5x.	 Vioxx-excluded screen first flags: 2006 one year <i>before</i> Nissen. Detection moved forward two years.
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Also from the timeline, same method, real receive dates: **Meridia × stroke flagged 2004** (n=16, PRR 5.0) — regulators acted 2009–10. **Darvon × cardiac arrest flagged 2005** (PRR 3.0) — EMA acted 2009, FDA 2010.

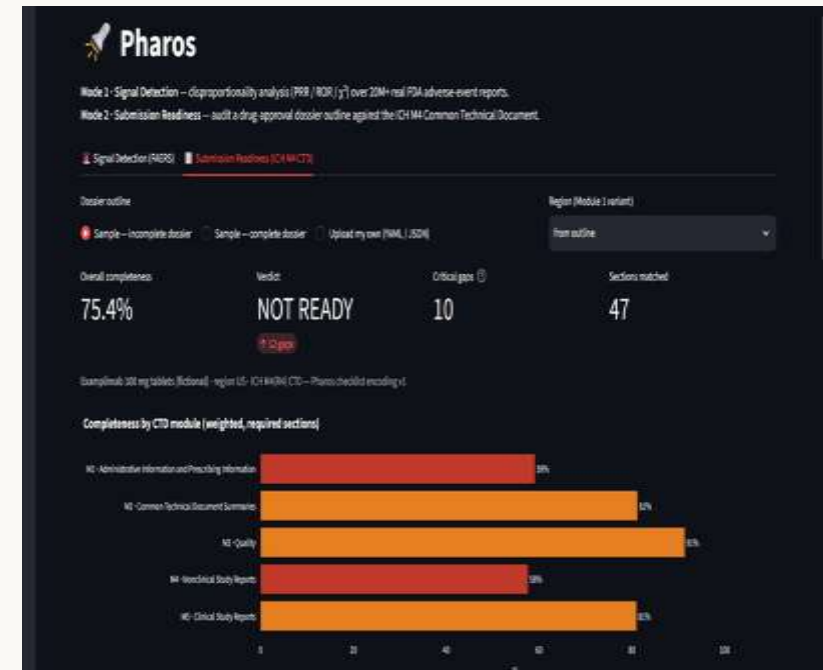
-  We expected Avandia to be obvious in 2005. It wasn't — and the tool told us why. Masking is in the literature (Maignen 2014); nobody puts it in the workflow. Pharos detects it every year and corrects it on request. That's the innovation: not a better number, a better question — "what hid this?"

Demo — Dashboard & CLI

Screenshots

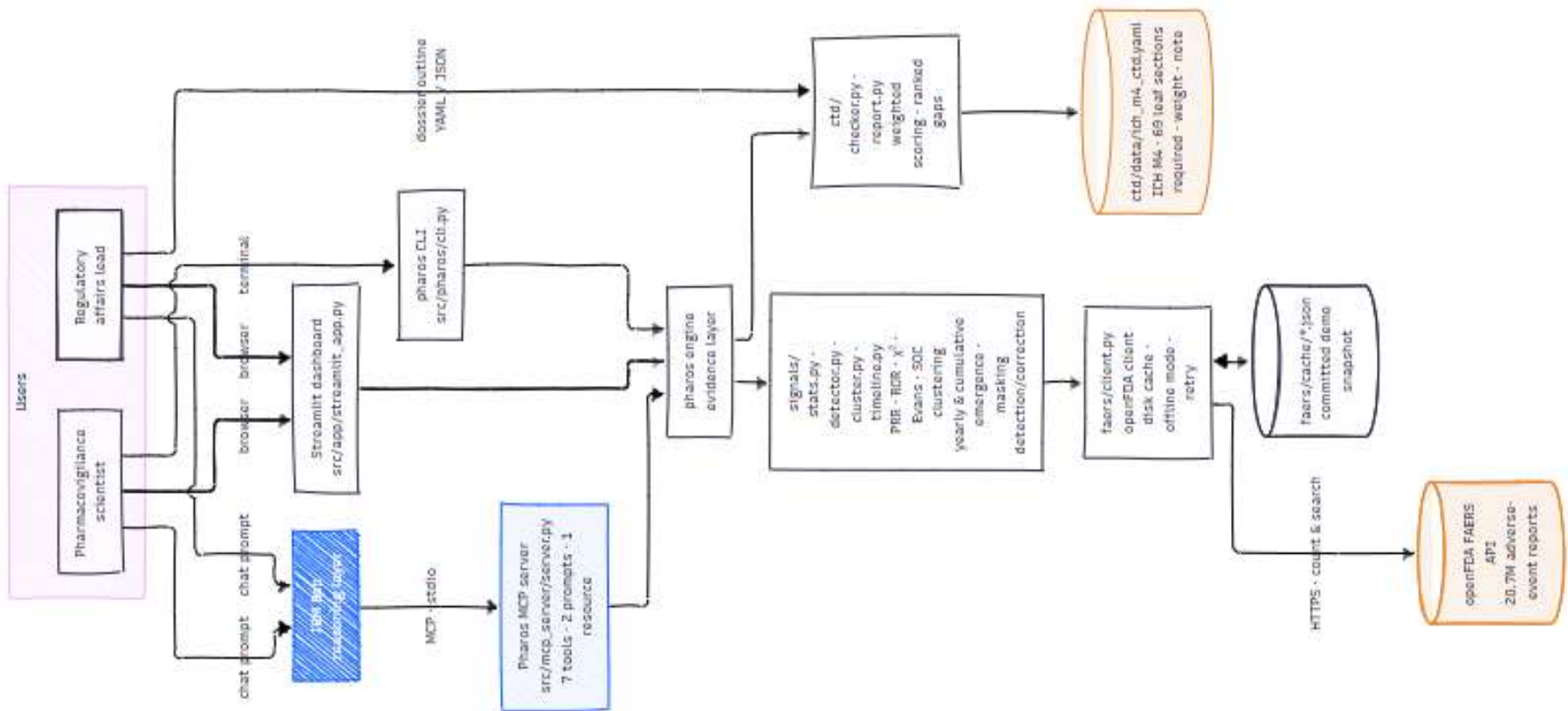


Mode 1: Vioxx: 35 clinical signals, cardiac cluster



Mode 2 : incomplete dossier: NOT READY, 10 critical gaps

Architecture



This is the template's own reference architecture — user → Bob → MCP → your server → data.

IBM Bob — Where and How (Load-Bearing)

Built with Bob

- Plan mode for the evidence/reasoning split
- Agent mode to implement
- /review on stats before tests
- subagent for openFDA research

Runs with Bob

`scan_signals` → `compute_prr` (verify) →
`search_reports` (ground) → `cluster_by_organ_system`
→ `signal_emergence_timeline` (date it; spot
masking; re-run corrected) → **memo**.

Server instructions: *always state n , PRR+CI, χ^2 ; name reporting biases; never claim causation.*

Remove Bob

→ no explanation, no conversation.

Remove Pharos

→ Bob is guessing.

Impact — What Changes

Signal detection

from periodic to continuous.

Dossier review

from "looks complete" to "here's the gap and why it matters."

Reviewer time

from reading everything to reading the right things.

Who benefits: PV scientists, regulatory leads, CROs, small biotechs without a \$2M PV platform.

What's next: E2B(R3) ICSR ingestion · EU EudraVigilance · automated Module 2.7.4 draft · Bob memo → Word/PDF.

Pharos doesn't replace the scientist. It replaces the pile.

Repo url : <https://github.com/Jainishcodee/bob-ai-hackathon-Sudo-Salingers>

